

Log-concavity of codimension-three level Hilbert functions of type two

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July 26, 2026

Abstract

Let $R = k[x_1, x_2, x_3]$, where k is any field. We prove that the Hilbert function $h = (1, 3, h_2, \dots, h_e = 2)$ of every standard graded Artinian level quotient of R of type two is log-concave. The argument uses the minimal resolution of the reindexed graded Matlis dual. Its last syzygy has rank one and full support; this forces every proper subset of the middle columns to be independent over $\text{Frac}(R)$. A putative failure of log-concavity then reduces to a rank-one family of relations, which determines a cyclic Gorenstein submodule. Stanley's characteristic-free theorem for codimension-three Gorenstein Hilbert functions closes the remaining case.

Introduction

Let A be a standard graded Artinian level algebra of codimension r and type t . Iarrobino [2] asked for which pairs (r, t) the Hilbert function of every such A is log-concave, and answered it positively for $r = 2$ and any t , and for $(r, t) = (3, 1)$, and negatively for $(r, t) = (4, 1)$. Zanello [5] settled every remaining pair negatively, with one exception: $(r, t) = (3, 2)$, which [5] records as open in every characteristic. Theorem 1 below resolves that case.

Zanello [5] raises the same question for pure O -sequences, equivalently for the Hilbert functions of *monomial* Artinian level algebras, where $(3, 2)$ was likewise the principal open case. That monomial case has been settled by a combinatorial argument [1]. It does not imply the result proved here: the Hilbert function of R/I is unchanged on passing to a generic initial ideal, but levelness and type are not preserved, since taking initial ideals can raise the socle type. The implication runs the other way, as every pure O -sequence of codimension three and type two is the Hilbert function of a monomial level algebra of codimension three and type two.

The proof below has been formalized in Lean 4 with Mathlib; the development is available at <https://github.com/tigerisaac/zanello-lc>. Its endpoint takes the level algebra and Lemma 1 as its only hypotheses, is free of `sorry`, and depends only on Lean's three standard foundational axioms.

Notation and the one imported theorem

For $j \geq 0$ set

$$N_j = \dim_k R_j = \binom{j+2}{2},$$

and put $N_j = 0$ for $j < 0$. We will use the following characteristic-free form of Stanley's theorem.

Lemma 1 (Stanley). *If B is a standard graded Artinian Gorenstein algebra of embedding dimension at most three and socle degree E , then its Hilbert function is symmetric and is nondecreasing through degree $\lfloor E/2 \rfloor$.*

More precisely, the Hilbert function is an SI-sequence [3]; the monotonicity stated above is the only part needed here. The result holds over an arbitrary field. A characteristic-free proof is given in [4].

The theorem

Theorem 1. *Let $A = R/I$ be a standard graded Artinian level algebra of embedding dimension three, socle degree e , and type two. Then*

$$h_i^2 \geq h_{i-1}h_{i+1} \quad (1 \leq i \leq e-1).$$

The statement holds in every characteristic.

Proof. Put $s = e + 3$.

The dual complex. Since A is Artinian, $\text{depth } A = 0$, so the graded Auslander–Buchsbaum formula gives $\text{pd}_R A = 3$ and the minimal free resolution of A has the form

$$0 \longrightarrow G_3 \longrightarrow G_2 \longrightarrow G_1 \longrightarrow R \longrightarrow A \longrightarrow 0.$$

Computing $\text{Tor}_3^R(A, k)$ from the Koszul resolution of k on x_1, x_2, x_3 gives

$$\text{Tor}_3^R(A, k) \cong \text{soc}(A)(-3),$$

while computing it from that resolution gives $G_3 \otimes_R k$. As A is level of type two with socle degree e , its socle is two-dimensional and concentrated in degree e , so

$$G_3 = R(-e-3)^2 = R(-s)^2.$$

Reindex the graded Matlis dual by setting

$$M_d = \text{Hom}_k(A_{e-d}, k), \quad g_d = \dim_k M_d = h_{e-d}.$$

By graded local duality $M \cong \text{Ext}_R^3(A, R(-s))$, so applying $\text{Hom}_R(-, R(-s))$ to that resolution yields a minimal exact complex

$$0 \longrightarrow R(-s) \xrightarrow{\delta_3} F_2 \xrightarrow{\delta_2} F_1 \xrightarrow{\delta_1} R^2 \longrightarrow M \longrightarrow 0, \quad (1)$$

where $F_i = \text{Hom}_R(G_{3-i}, R(-s))$ and the two outer terms are $\text{Hom}_R(R, R(-s)) = R(-s)$ and $\text{Hom}_R(R(-s)^2, R(-s)) = R^2$.

Thus M is generated by two elements in degree zero. Its socle is one-dimensional and concentrated in degree e . Indeed, for $d < e$, standard generation gives $A_{e-d} = R_1 A_{e-d-1}$, so no nonzero functional in M_d is annihilated by all of R_1 .

Write

$$F_1 = \bigoplus_j R(-j)^{p_j}, \quad F_2 = \bigoplus_j R(-j)^{q_j}.$$

The circuit. Let $K = \text{Frac}(R)$. Tensoring (1) with K shows that the kernel of δ_2 is one-dimensional. Its spanning vector has no zero coordinate. Indeed, $\delta_3 = \text{Hom}_R(\varepsilon, R(-s))$ where $\varepsilon: G_1 \rightarrow R$ is the first differential of the resolution of A , whose entries are a minimal homogeneous generating set of I ; dualizing, the image of a generator of $R(-s)$ under δ_3 has those same entries as its coordinates, and a minimal generator of a nonzero ideal is nonzero. Consequently,

$$\text{every proper subset of the columns of } \delta_2 \text{ is } K\text{-linearly independent.} \quad (2)$$

The rank estimate. Fix d with $2 \leq d \leq e$. The corresponding log-concavity inequality is the one centered at g_{d-1} . Define

$$Q_d = \sum_{j \leq d} q_j, \quad P_{<d} = \sum_{j < d} p_j,$$

and let r_d be the K -rank of the restriction of δ_1 to the summands of F_1 having shift strictly less than d . Thus $r_d \in \{0, 1, 2\}$. Finally, set

$$\varepsilon_d = \begin{cases} 1, & Q_d = \text{rank } F_2, \\ 0, & Q_d < \text{rank } F_2. \end{cases}$$

By (2), the columns of δ_2 having shift at most d have rank $Q_d - \varepsilon_d$. Since every differential in a minimal graded free resolution has entries in the homogeneous maximal ideal, these columns can have nonzero entries only in rows of F_1 whose shifts are strictly less than d . Their images lie in the kernel of a rank- r_d map from a $P_{<d}$ -dimensional K -space to K^2 . Therefore

$$Q_d - \varepsilon_d \leq P_{<d} - r_d. \quad (3)$$

Let $\Delta^2 g_d = g_d - 2g_{d-1} + g_{d-2}$. The Hilbert numerator of (1) is

$$(1-t)^3 \sum_{j \geq 0} g_j t^j = 2 - \sum_j p_j t^j + \sum_j q_j t^j - t^s.$$

Since $d \leq e < s$, summing its coefficients through degree d gives

$$\Delta^2 g_d = 2 + Q_d - P_{<d} - p_d.$$

Together with (3), this yields the central estimate

$$\Delta^2 g_d \leq 2 - r_d + \varepsilon_d - p_d. \quad (4)$$

If $\Delta^2 g_d \leq 0$, then $g_{d-2} + g_d \leq 2g_{d-1}$, and the arithmetic-geometric mean inequality gives $g_{d-2}g_d \leq g_{d-1}^2$. Thus a failure of log-concavity would imply

$$\Delta^2 g_d \geq 1. \quad (5)$$

Case analysis. First suppose $\varepsilon_d = 1$. Every summand $R(-b)$ of F_2 then has $b \leq d$. By the degree correspondence in the dual resolution, it comes from a minimal generator of I of degree $s-b \geq s-d$. Thus I has no nonzero element in degrees below $s-d$. With $m = e-d+2 = s-d-1$, the tested triple is

$$(g_{d-2}, g_{d-1}, g_d) = (N_m, N_{m-1}, N_{m-2}).$$

Since the ratios N_j/N_{j-1} are nonincreasing, this triple is log-concave.

We may therefore assume $\varepsilon_d = 0$. If $r_d = 2$, (4) gives $\Delta^2 g_d \leq 0$, contradicting (5).

If $r_d = 0$, there are no summands of F_1 of shift below d . Indeed, each such summand gives a nonzero minimal relation, so the restricted map would have positive rank otherwise. Equation (3) then gives $Q_d = 0$. Consequently

$$g_{d-2} = 2N_{d-2} = d(d-1), \quad g_{d-1} = 2N_{d-1} = d(d+1),$$

and

$$g_d = 2N_d - p_d = (d+1)(d+2) - p_d.$$

A direct calculation gives

$$g_{d-1}^2 - g_{d-2}g_d = d(2(d+1) + (d-1)p_d) > 0.$$

The critical case $r_d = 1$. From (4) and (5), a failure can occur only if

$$p_d = 0, \quad \Delta^2 g_d = 1. \tag{6}$$

All columns of δ_1 having shift below d span one line over K . Because R is a UFD, this line has a primitive homogeneous generator $v = (v_1, v_2) \in R^2$, whose two components have a common degree $a < d$. The submodule generated by the low-shift columns is then vJ for a homogeneous ideal $J \subseteq R$. To see this, divide one nonzero column by the gcd of its two entries. The resulting primitive vector spans the line, and Euclid's lemma shows that every other column is an R -multiple of it.

Because $p_d = 0$, no relation has shift d , while relations of shift greater than d have no component in degrees at most d . Therefore

$$(\ker(R^2 \rightarrow M))_t = (vJ)_t \quad (t \leq d). \tag{7}$$

Let H be the image of v in M_a and put $n = d - a$.

If $H = 0$, then v itself is a relation of degree $a < d$. Equation (7) and the primitivity of v force $1 \in J$, so $J = R$. In this case set $B_j = 0$ for every j .

Suppose $H \neq 0$. The finite-length module RH has nonzero socle, and this socle is contained in the one-dimensional socle of M . Consequently

$$B = R/\text{Ann}(H)$$

is Artinian Gorenstein, has embedding dimension at most three, and has socle degree $E = e - a$. We write $B_j = \dim_k(R/\text{Ann}(H))_j$ for its Hilbert function. For every $j \leq n$ and $c \in R_j$,

$$cH = 0 \iff cv \in \ker(R^2 \rightarrow M)_{a+j} \stackrel{(7)}{\iff} cv \in (vJ)_{a+j} \iff c \in J_j.$$

The last equivalence uses the injectivity of multiplication by the nonzero vector v . Thus J and $\text{Ann}(H)$ agree through degree n .

Equation (8). In both the $H = 0$ and $H \neq 0$ cases, with negative subscripts interpreted as zero, we claim that for $t = d - 2, d - 1, d$,

$$g_t = 2N_t - N_{t-a} + B_{t-a}. \quad (8)$$

Indeed, M is the cokernel of δ_1 , so counting dimensions in degree t gives

$$g_t = 2N_t - \dim_k(\ker(R^2 \rightarrow M))_t.$$

For $t \leq d$ that kernel is $(vJ)_t$ by (7). Multiplication by v is an injective map $R \rightarrow R^2$, homogeneous of degree a , since $v \neq 0$ and R is a domain; hence $(vJ)_t \cong J_{t-a}$ and $\dim_k(vJ)_t = \dim_k J_{t-a}$. Finally $\dim_k J_{t-a} = N_{t-a} - \dim_k(R/J)_{t-a}$, and for $t \leq d$ we have $t - a \leq n$, so $\dim_k(R/J)_{t-a} = B_{t-a}$ because J and $\text{Ann}(H)$ agree through degree n . Combining these gives (8).

Conclusion. Equation (8) gives $g_{d-1} \geq N_{d-1}$, since $B_{d-1-a} \geq 0$ and $N_{d-1-a} \leq N_{d-1}$. On the other hand,

$$g_{d-1} = h_{e-d+1} \leq N_{e-d+1}.$$

Since N_j is strictly increasing, it follows that

$$2d \leq e + 2. \quad (9)$$

When $H \neq 0$, this gives

$$2(n-1) = 2d - 2a - 2 \leq e - 2a \leq e - a = E,$$

so Lemma 1 applies to B at the indices $n-2 \leq n-1$ and gives

$$D := B_{n-1} - B_{n-2} \geq 0.$$

(If $n \leq 1$ then $B_{n-2} = 0$ by the convention on negative subscripts, and $D \geq 0$ holds because $B_{n-1} \geq 0$, without appeal to Lemma 1.) For $H = 0$ the inequality $D \geq 0$ also holds, since $D = 0$ by definition.

Set $x = g_{d-1} - g_{d-2}$. Using (8) and $N_j - N_{j-1} = j + 1$, we obtain

$$x = 2d - n + D = d + a + D \geq d.$$

Also $g_{d-2} \leq 2N_{d-2} = d(d-1)$.

Finally, (6) says that the tested triple has the form

$$(g_{d-2}, g_{d-1}, g_d) = (y, y + x, y + 2x + 1)$$

for $y = g_{d-2}$. Its log-concavity margin is

$$g_{d-1}^2 - g_{d-2}g_d = (y+x)^2 - y(y+2x+1) = x^2 - y \geq d^2 - d(d-1) = d > 0.$$

This contradicts the assumed failure.

We have proved log-concavity of g at every index $d-1$ with $2 \leq d \leq e$. Reversal preserves the log-concavity inequalities, and $g_d = h_{e-d}$, so the same inequalities hold for h . $\square$

Remark. The argument is valid over every field. The substantive theorem invoked is the characteristic-free form of Stanley's theorem in Lemma 1; the remaining steps use graded Matlis duality, linear algebra over $\text{Frac}(R)$, and the UFD property of $k[x_1, x_2, x_3]$.

Acknowledgment

Artificial intelligence was used in producing both the proof and its Lean formalization.

References

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2020 Mathematics Subject Classification. Primary 13D40; Secondary 13H10, 13D02, 05E40.

Key words and phrases. Level algebra, Hilbert function, log-concavity, minimal free resolution, graded Matlis duality, Gorenstein algebra.